

Contact

[Your phone]

ruyangearnold@gmail.com

[Your city, country]

arnold.dev

GitHub

SKILLS

Frontend

React

Next.js

TypeScript

Astro

Tailwind CSS

Backend

Node.js

Python

Java

PostgreSQL

MongoDB

Mobile

React Native

Expo

Cross-platform UI

Mobile APIs

AI / ML

TensorFlow

CNN

Machine Learning

Python

Data Preprocessing

DevOps

Docker

AWS

GitHub Actions

MQTT / IoT

Security

Cybersecurity

Penetration Testing

Network Security

OWASP

LANGUAGES

English (Fluent)

[Add language]

CERTIFICATIONS

Cybersecurity Fundamentals

Self-Directed Study · 2024

Full-Stack & AI Project Portfolio

Project-Based Learning · 2025

RUYANGE ARNOLD

FULL-STACK DEVELOPER

PROFILE

Full-stack developer with 3+ years of hands-on experience building Java, Python, AI/ML, and web applications. I design and ship end-to-end systems — from desktop GUIs and REST APIs to CNN models and IoT pipelines.

Portfolio of RUYANGE Arnold — focused on practical software engineering through open-source work on GitHub, project-based learning, and production-ready user experiences.

WORK EXPERIENCE

University / Self-Directed Learning 2023 - Present

Computer Science Student

Studying computer science fundamentals while building personal projects in Java, Python, AI/ML, and web development. Focus on practical software engineering through open-source work on GitHub.

Java

Python

Next.js

TensorFlow

SQL

Personal Projects & Open Source 2023 - Present

Independent Developer

Designed and shipped projects including an e-commerce platform (arnshop), handwriting recognition CNN, MQTT IoT weather system, web scraping pipelines, and this portfolio site.

Java

Python

Flask

Docker

Machine Learning

EDUCATION

Computer Science 2023 - Present

University / Self-Directed Learning · Remote

Studying computer science fundamentals while building personal projects in Java, Python, AI/ML, and web development. Focus on practical software engineering through open-source work.

FEATURED PROJECTS

Selected work across full-stack, backend, AI/ML, DevOps, and mobile-focused development. Each project includes the problem, implementation, outcome, and repository link.

FULLSTACK PROJECTS

arnshop — E-Commerce Platform

Problem: Needed a complete desktop e-commerce system for selling electronics with inventory and checkout — without relying on third-party storefronts.

Built: A Java Swing application with MySQL backend: product catalog, shopping cart, order management, and admin inventory controls.

Outcome: Fully functional e-commerce workflow with clean Java architecture and a production-ready desktop UI.

Tech: Java, Swing, MySQL, E-Commerce

GitHub: <https://github.com/Arn-The-Wolf/arnshop>

Key results

Full e-commerce functionality

Clean Java architecture

Production-ready UI

Challenges solved

Building intuitive GUI

Inventory management

Payment flow integration

Impact metrics

Features: 0 → Full cart + checkout (100%)

GUI Screens: N/A → Multi-panel UI (Complete)

Data Layer: None → MySQL integrated (Production-ready)

arnlibrary — Library Management System

Problem: Small libraries needed a simple system to track books, students, and borrowing without complex enterprise software.

Built: Java Swing GUI with MySQL persistence for books, students, due dates, and admin operations.

Outcome: End-to-end library workflow with CRUD operations and an interactive admin panel.

Tech: Java, Swing, MySQL, GUI

GitHub: <https://github.com/Arn-The-Wolf/arnlibrary>

Key results

Full library workflow

Interactive admin panel

Challenges solved

CRUD operations

GUI design

Data persistence

arnold.dev Portfolio — Personal Developer Portfolio

Problem: Needed a professional online presence to showcase projects, skills, and case studies with fast performance and a distinctive modern UX.

Built: Full-stack portfolio with Next.js 14, TypeScript, Tailwind CSS, Three.js visuals, JSON-backed CMS APIs, and Vercel deployment.

Outcome: Live portfolio at arnold.dev with project showcase, skills arsenal, blog, case studies, and contact flow.

Tech: Next.js, TypeScript, Tailwind CSS, React, Three.js, Vercel

GitHub: <https://github.com/Arn-The-Wolf>

Key results

Live production site
CMS-driven content
Performance-optimized UX
Challenges solved
3D hero performance
Content management
Responsive design

BACKEND PROJECTS

webscraping — Python Data Extraction

Problem: Manual data collection from multi-page websites was slow and error-prone for analysis workflows.

Built: A Python pipeline using BeautifulSoup to crawl pages, parse HTML, normalize records, and export structured datasets.

Outcome: Automated extraction pipeline producing clean structured output suitable for downstream analysis.

Tech: Python, BeautifulSoup, Web Scraping, Data Extraction

GitHub: <https://github.com/Arn-The-Wolf/webscraping>

Key results

Automated data pipeline

Clean structured output

Challenges solved

Multi-page navigation

Data structuring

Rate limiting

BFS-DFS Algorithms — Graph Traversal Implementation

Problem: Understanding and comparing graph traversal strategies for pathfinding and connected-component problems.

Built: Java implementations of BFS and DFS with graph representation, traversal logic, and performance comparison.

Outcome: Correct, tested traversal implementations demonstrating algorithm trade-offs.

Tech: Java, Algorithms, Data Structures

GitHub: <https://github.com/Arn-The-Wolf/Breadth-First-Searching-Deep-First-Searching->

Key results

Working BFS/DFS implementations

Challenges solved

Algorithm optimization

Graph representation

AI / ML PROJECTS

Handwriting Recognition AI — CNN Machine Learning Model

Problem: Classifying handwritten digits and text manually does not scale — needed a trained model for automated recognition.

Built: A CNN in Python with TensorFlow: dataset preprocessing, model training, evaluation metrics, and inference on new samples.

Outcome: Working classifier with extensible training pipeline and documented accuracy improvements across iterations.

Tech: Python, TensorFlow, CNN, Machine Learning, AI

GitHub: <https://github.com/Arn-The-Wolf/Handwriting-Recognition-Ai>

Key results

Working CNN model

Digit classification

Extensible architecture

Challenges solved

Model training optimization

Dataset preprocessing

Accuracy improvement

Impact metrics

Model Type: None → CNN classifier (Deployed)

Dataset Handling: Raw → Preprocessed (Optimized)

Accuracy: Baseline → High confidence (Strong)

DEVOPS PROJECTS

mqtt-weather-app — IoT Weather Monitoring

Problem: Weather sensor data was scattered across devices with no unified real-time view or reliable ingestion path.

Built: Flask API subscribing to MQTT topics, processing sensor payloads, and exposing a live weather dashboard.

Dockerized for deployment.

Outcome: Live MQTT pub/sub pipeline with a working dashboard for environmental monitoring.

Tech: Python, Flask, MQTT, IoT, Docker

GitHub: <https://github.com/Arn-The-Wolf/mqtt-weather-app>

Key results

Live weather dashboard

MQTT pub/sub pipeline

Challenges solved

MQTT broker integration

Real-time data processing

Sensor calibration

Impact metrics

Data Pipeline: Manual → MQTT live stream (Real-time)

Backend: None → Flask + MQTT (Complete)

Deployment: Local → Dockerized (Portable)

MOBILE — SKILLS & FOCUS

Mobile development is an active focus area. Current stack centers on React Native and Expo for cross-platform apps, with API-first backends from my full-stack work.

Technologies

React Native

Expo

Cross-platform UI

Mobile APIs

No dedicated mobile app repositories in the portfolio yet — upcoming work will extend existing backends (Flask, Node.js) with React Native clients.